

Applied Financial Risk Management

Session 10

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Volatility forecast: Beyond RiskMetrics

- RiskMetrics forecast for t+1 variance can be written as

$$\sigma_{t+1}^2 = \lambda \sigma_t^2 + (1 - \lambda) R_t^2$$
- As we saw earlier, RiskMetrics sets $\lambda = 0.94$, for all asset classes
 - You don't like such "dirty" ad-hoc estimates?
- The above expression for volatility forecast can be written more generally as

$$\sigma_{t+1}^2 = \omega + \alpha R_t^2 + \beta \sigma_t^2$$
 - If we set $\omega=0$, $\alpha = 1 - \lambda$, $\beta = \lambda$, then we get the RiskMetrics model
- The generic model is the simplest GARCH model
 - GARCH refers to Generalized Autoregressive Conditional Heteroskedasticity
 - The model that we have shown is commonly referred to as GARCH(1,1) model

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Volatility forecast: GARCH Estimation

- The GARCH (1,1) model can be summarized as:

$$R_{t+1} = \sigma_{t+1}\varepsilon_{t+1}, \varepsilon_{t+1} \sim N(0,1)$$

$$\sigma_{t+1}^2 = \omega + \alpha R_t^2 + \beta \sigma_t^2$$

- The returns are conditionally normally distributed, i.e., return for period $t+1$ follows normal distribution with mean 0 and standard deviation σ_{t+1} .
 - Note, σ_{t+1} is based on information available up to time t .
 - It is also dependent on the three parameters : ω, α, β
- Estimate the GARCH model for the period under consideration using R.

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- `rm(list=ls())`
- `setwd("C:\\")`
- `nifty=read.table("Nifty_data.csv", header=T, sep=",")`
- `library(rugarch)`
- `spec1=ugarchspec(variance.model = list(model = "sGARCH",
garchOrder = c(1, 1)),mean.model = list(armaOrder = c(0, 0),
include.mean = FALSE))`
- `garch_fit_1 = ugarchfit(spec = spec1, data = nifty)`
- `Sigma=garch_fit_1@fit$sigma`

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rugarch

- **ugarchspec:** For specifying GARCH family of models

```
ugarchspec(variance.model = list(model = "sGARCH", garchOrder = c(1, 1),
submodel = NULL, external.regressors = NULL, variance.targeting = FALSE),
mean.model = list(armaOrder = c(1, 1), include.mean = TRUE, archm = FALSE,
archpow = 1, arfima = FALSE, external.regressors = NULL, archex = FALSE),
distribution.model = "norm", start.pars = list(), fixed.pars = list(), ...)
```

- Variance models: sGARCH, eGARCH, gjrGARCH
- Distribution.model: "norm","snorm","std","sstd","ged" etc.

- **ugarchfit:** For fitting the specified GARCH model

```
ugarchfit(spec, data, out.sample = 0, solver = "solnp", solver.control = list(),
fit.control = list(stationarity = 1, fixed.se = 0, scale = 0, rec.init = 'all'),
numderiv.control = list(grad.eps=1e-4, grad.d=0.0001,
grad.zero.tol=sqrt(.Machine$double.eps/7e-7), hess.eps=1e-4, hess.d=0.1,
hess.zero.tol=sqrt(.Machine$double.eps/7e-7), r=4, v=2),...)
```

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The GJR-GARCH model

- Due to Glosten, Jaganathan and Runkle

$$\sigma_t^2 = \alpha_0 + \alpha_1 u_{t-1}^2 + \beta \sigma_{t-1}^2 + \gamma u_{t-1}^2 I_{t-1}$$

where $I_{t-1} = 1$ if $u_{t-1} < 0$
 $= 0$ otherwise

- For a leverage effect, we would see $\gamma > 0$.
- We require $\alpha_1 + \gamma \geq 0$ and $\alpha_1 \geq 0$ for non-negativity.

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EGARCH model

- Suggested by Nelson (1991). The variance equation is given by

$$\log(\sigma_t^2) = \omega + \beta \log(\sigma_{t-1}^2) + \gamma \frac{u_{t-1}}{\sqrt{\sigma_{t-1}^2}} + \alpha \left[\frac{|u_{t-1}|}{\sqrt{\sigma_{t-1}^2}} - \sqrt{\frac{2}{\pi}} \right]$$

- Advantages of the model
 - Since we model the $\log(\sigma_t^2)$, then even if the parameters are negative, σ_t^2 will be positive.
 - We can account for the leverage effect: if the relationship between volatility and returns is negative, γ , will be negative